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1*H*-1,2,4-Triazolo-5-diazonium Salts in the Synthesis of New 6*H*-Benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-ones and [1,2,4]Triazolo[5',1':3,4][1,2,4]triazino-[6,5-*c*]quinolin-6(7*H*)-ones

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Abstract—A convenient approach to the synthesis of new 2-*R*-6*H*-benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*]-[1,2,4]triazin-6-ones and 2-*R*-[1,2,4]triazolo[5',1':3,4][1,2,4]triazino[6,5-*c*]quinolin-6(7*H*)-ones, based on the condensation of 3-*R*-1*H*-1,2,4-triazolo-5-diazonium salts with 2*H*-1-benzopyran-2,4(3*H*)-dione and quinoline-2,4(1*H*,3*H*)-dione, is developed.

Keywords: 1*H*-1,2,4-triazolo-5-diazonium salt, 2*H*-1-benzopyran-2,4(3*H*)-dione, quinoline-2,4(1*H*,3*H*)-dione, hydrazones, 6*H*-benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-ones, -[1,2,4]triazolo[5',1':3,4][1,2,4]-triazino[6,5-*c*]quinolin-6(7*H*)-ones

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INTRODUCTION

One of the modern concepts of the design of heterocyclic systems in the context of searching for new biologically active compounds includes the strategy of combining several pharmacophore fragments in one molecule [1–8]. The implementation of this approach using substructures that are part of already known drugs seems very promising [9, 10].

2-Quinone derivatives have found application as antibacterial [11], anticancer [12], antituberculosis [13], and antihepatitis agents [14]. In addition, some dihydroquinones [15, 16] and coumarins [17–19] exhibit high inhibitory activity in relation to the blood clotting factors Xa and XIa. In turn, the 1,2,4-triazolo[5,1-*c*]-[1,2,4]triazine system is interesting in that its derivatives obtained by the condensation of 1*H*-1,2,4-triazolo-5-diazonium salts with methylene-active nitriles, enaminones, and enols of heterocyclic ketones, were found to act as antibacterial [20, 21] and antiviral agents [22–26]. An example of such compounds is provided by Triazavirin, a drug introduced into clinical practice.

In the search for new hybrid molecules containing antiviral and anticoagulant fragments, we proposed to combine the 2-quinone, 1-benzopyran-2-one, and 1,2,4-triazolo[5,1-*c*][1,2,4]triazine scaffolds (Fig. 1). Moreover, these compounds can exhibit chelating

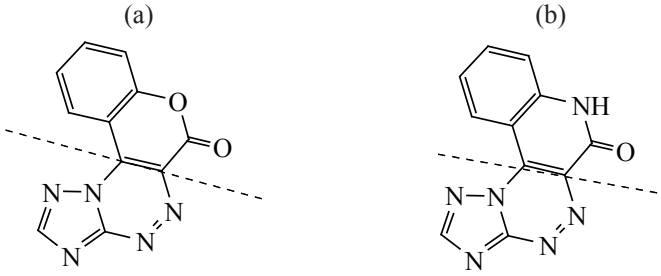


Fig. 1. Hybride molecules (a) 2-R-6*H*-[1]benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-one and (b) 2-*R*-[1,2,4]triazolo-[5',1':3,4][1,2,4]triazino[6,5-*c*]quinoline-6(7*H*)-one.

properties and find application in microelectronics for chemical and electrochemical metal deposition.

RESULTS AND DISCUSSION

Recently Jashari et al. [27] reported an example of the combination of 1*H*-1,2,4-triazolo-5-diazonium salt with 2*H*-1-benzopyran-2,4(3*H*)dione, the product of which was assigned the structure of 3-[2-(1*H*-1,2,4-triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-dione. We established that the combination of 3-*R*-1*H*-1,2,4-triazolo-5-diazonium salts **2a–2d**, obtained by the diazotization of 3-*R*-5-amino-1*H*-1,2,4-triazolos **1a–1d** in the presence of sulfuric acid, with 2*H*-1-benzopyran-2,4(3*H*)-dione (**3**) leads to the formation of compounds **4a–4d** (Scheme 1).

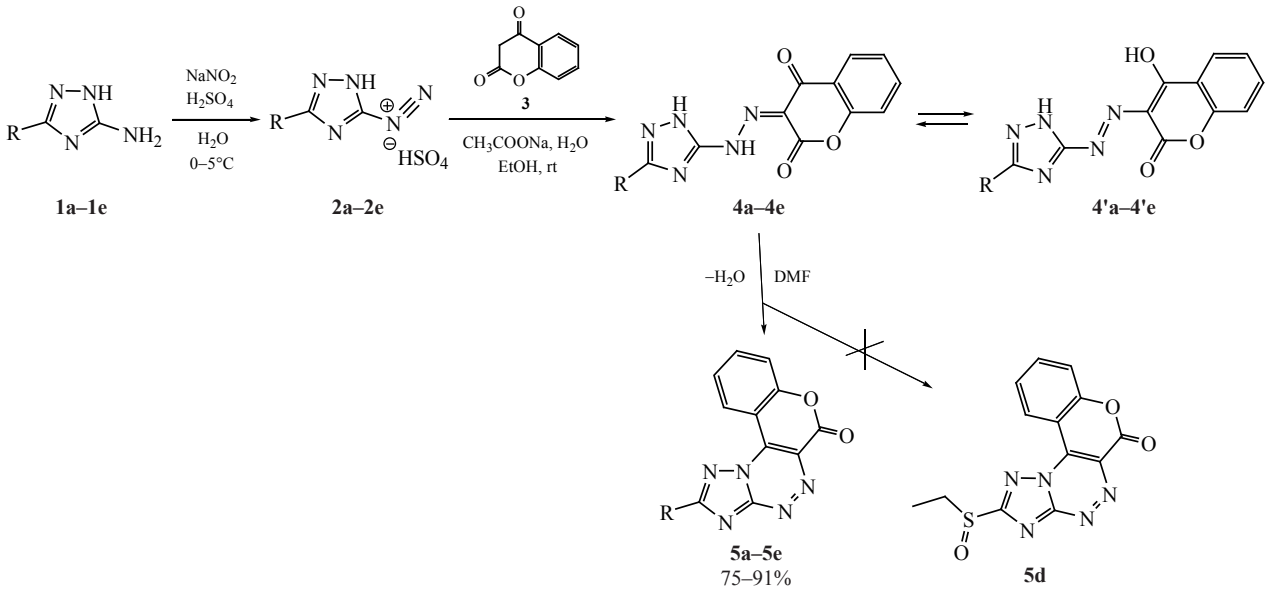
HPLC–MS analysis of the reaction products showed that they are individual compounds with the molecular ions $[M + H]^+$ corresponding to the structure of 3-[2-(3-*R*-1*H*-1,2,4-triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-diones. At the same time, the ^{1}H NMR spectra of compounds **4** in DMSO-d_6 display complex sets of proton signals, which can be explained by the presence of diazenyl tautomers **4'a–4'd**.

A short reflux of hydrazones **4a–4c** in DMF resulted in the exclusive isolation of cyclization products 2-*R*-6*H*-benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]-triazine-6-ones **5a–5c** (Scheme 1). The ^{1}H NMR spectra of these products contained no NH proton signals of the triazole ring and the hydrazone fragment and showed proton signals of the benzene ring and the corresponding substituents in the triazole ring. Regretfully, we failed to obtain the cyclization product of hydrazone **4d**. Even after refluxing for many hours in DMF, no 2-(ethylsulfinyl)-6*H*-benzopyrano[4,3-*e*]-[1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-one (**5d**) formed. This result can be explained by the electron-acceptor properties of the sulfinyl group, which prevents the final stage of intramolecular cyclization of hydrazone **4d**.

By coupling quinoline-2,4(1*H*,3*H*)-dione (**6**) with triazolo-5-diazonium salts **2a–2c** and **2e**, we synthesized hydrazones **7a–7d**. According to HPLC-MS, the products are individual compounds with the molecular ions $[M + H]^+$ corresponding to the structure 3-[2-(3-*R*-1*H*-1,2,4-triazol-5-yl)hydrazono]quinoline-2,4(1*H*,3*H*)-diones. The ^{1}H NMR spectra of hydrazones **7a–7d** show ever more sets of proton signals than the spectra of hydrazones **4a–4d**, presumably due to the presence of two diazenyl tautomers **7'a–7'd** and **7''a–7''d** (Scheme 2).

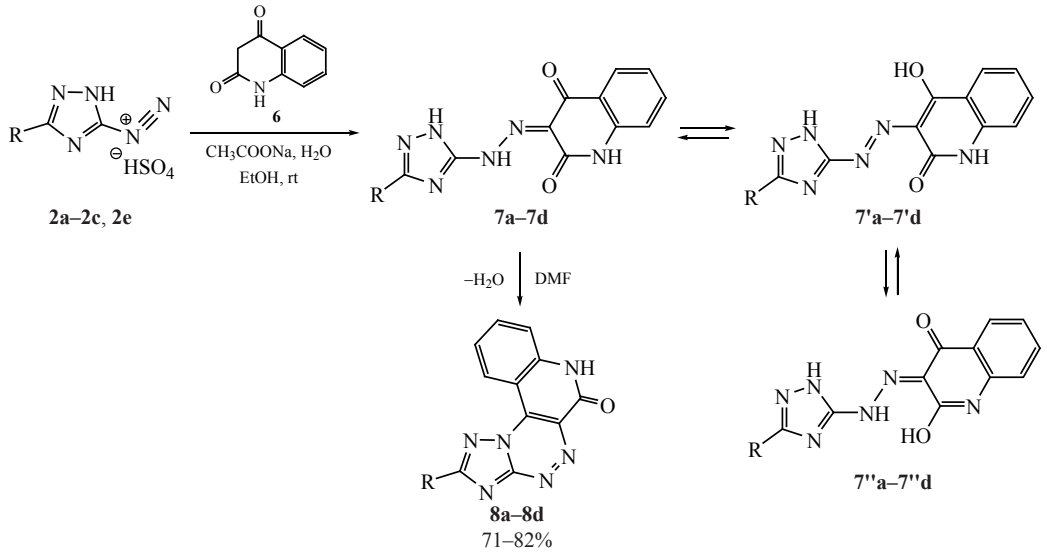
Refluxing hydrazones **7a–7d** in DMF with the aim to study their possible intramolecular cyclization gave previously unknown 2-*R*-[1,2,4]triazolo[5',1':3,4]-[1,2,4]triazino[6,5-*c*]quinoline-6(7*H*)-ones **8a–8d** with yields of 71–82% (Scheme 2). The ^{1}H NMR spectra of the synthesized triazolotriazinoquinolines **8** do not contain signals from the NH protons of the triazole ring

Scheme 1.



R = H (**a**), CH_3 (**b**), CH_3S (**c**), C_2H_5SO (**d**).

Scheme 2.



R = H (**a**), CH_3 (**b**), CH_3S (**c**), C_2H_5 (**d**).

and the hydrazone fragment but show protons signals of the benzene ring as 2 multiplets in the regions of ~7.55 and 8.55 ppm and a doublet at 9.51–9.66 ppm. The quinoline NH proton signal is observed in the range of 12.49–12.66 ppm.

EXPERIMENTAL

The ^{1}H NMR spectra were registered on a Bruker DRX-500 spectrometer at 500 MHz and an Agilent MR 400+ spectrometer at 400 MHz for DMSO-d_6 solutions, internal standard TMS. The HPLC–MS analysis was performed on an Agilent Technologies 1260 Infinity Binary LC coupled with an Agilent 6230 Accurate-Mass TOF LC/MS mass detector and a dual ESI ion source. The spectra were recorded in a positive ion mode; nebulizer gas (N_2) 20 psig, drying gas (N_2) 6 mL/min, 325°C; mass range 50–2000 Da; voltages: capillary 4.0 kV, fragmenter +191 V, skimmer +66 V, and OctRF 750 V. Cromatographic separation was performed using a Poroshell 120 EC-C18 (4.6 × 50 mm; 2.7 μm), gradient elution with acetonitrile–water (0.1% formic acid), flow rate 0.4 mL/min. The results were processed using an Agilent MassHunter Workstation Data Acquisition Software (version B.06.00). The melting points were determined using a Stuart SMP30 melting point apparatus. The purity of the starting and synthesized compounds, as well as reaction progress were monitored by TLC on Merck TLC Silica gel 60 F_{254} in $CHCl_3$–MeOH, 6 : 1 (spot visualization under UV light).

The starting 3-*R*-5-amino-1*H*-1,2,4-triazoles **1a–1e** were purchased from Alinda Chemical.

3-[2-(3-*R*-1*H*-1,2,4-Triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-diones 4a–4d and 3-[2-(3-R-1*H*-1,2,4-triazol-5-yl)hydrazono]quinoline-2,4-(1*H*,3*H*)-diones 7a–7d (*general procedure*). To a stirred mixture of 5 mmol of 3-*R*-5-amino-1*H*-1,2,4-triazole **1a–1e**, 5 mL of H_2O, and 0.4 mL of conc. H_2SO_4, cooled to 0°C, a 5% aqueous solution of 5 mmol of $NaNO_2$ was added dropwise at such rate that the temperature of the reaction mixture is maintained at no more than 5°C. The solution was left to stand at this temperature for 5 min and then added in portions to a mixture of 0.81 g (5 mmol) of 2*H*-chromene-2,4(3*H*)-dione (**3**) or 0.81 g (5 mmol) of quinoline-2,4(1*H*,3*H*)-dione (**6**), 5 g of AcONa, 2 mL of EtOH, and 40 mL of H_2O, cooled to 10°C. The reaction mixture was stirred for 3 h, and the precipitate that formed was filtered off, washed with water, and dried. The products were further used without additional purification.

3-[2-(1*H*-1,2,4-Triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-dione (4a) was prepared using 0.42 g of compound **1a**. Yield 1.26 g (98%), orange crystals, mp 225–228°C (225–227°C [27]). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 258.0626 (100) $[M + H]^+$. $C_{11}H_7N_5O_3^+$. *M* 258.0622.

3-[2-(3-Methyl-1*H*-1,2,4-triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-dione (4b) was prepared using 0.49 g of compound **1b**. Yield 1.33 g (97%), orange crystals, mp 233–236°C. Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 272.0781 (100) $[M + H]^+$. $C_{12}H_9N_5O_3^+$. *M* 272.0779.

3-[2-(3-Methylsulfanyl-1*H*-1,2,4-triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-dione (4c) was prepared using 0.65 g of compound **1c**. Yield 1.45 g (96%), orange crystals, mp 242–245°C. Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 304.0500 (100) $[M + H]^+$. $C_{12}H_9N_5O_3S^+$. *M* 304.0499.

3-[2-(3-Ethylsulfinyl-1*H*-1,2,4-triazol-5-yl)hydrazono]-2*H*-1-benzopyran-2,4(3*H*)-dione (4d) was prepared using 0.80 g of compound **1d**. Yield 1.59 g (95%), orange crystals, mp 247–251°C. Mass spectrum, *m/z* (I_{rel}, %): 334.0601 (100) $[M + H]^+$. $C_{13}H_{11}N_5O_4S^+$. *M* 334.0605.

3-[2-(1*H*-1,2,4-Triazol-5-yl)hydrazono]quinoline-2,4(1*H*,3*H*)-dione (7a) was prepared using 0.42 g of compound **1a**. Yield 1.23 g (96%), orange crystals, mp > 300°C (decomp.). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 257.0780 (100) $[M + H]^+$. $C_{11}H_8N_6O_2^+$. *M* 257.0782.

3-[2-(3-Methyl-1*H*-1,2,4-triazol-5-yl)hydrazono]-quinoline-2,4(1*H*,3*H*)-dione (7b) was prepared using 0.49 g of compound **1b**. Yield 1.33 g (98%), orange crystals, mp > 300°C (decomp.). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 271.0937 (100) $[M + H]^+$. $C_{12}H_{10}N_6O_2^+$. *M* 271.0938.

3-[2-(3-Methylsulfanyl-1*H*-1,2,4-triazol-5-yl)hydrazono]quinoline-2,4(1*H*,3*H*)-dione (7c) was prepared using 0.65 g of compound **1c**. Yield 1.44 g (95%), mp > 300°C (decomp.). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 303.0655 (100) $[M + H]^+$. $C_{12}H_{10}N_6O_2S^+$. *M* 303.0659.

3-[2-(3-Ethyl-1*H*-1,2,4-triazol-5-yl)hydrazono]-quinoline-2,4(1*H*,3*H*)-dione (7d) was prepared using 0.56 g of compound **1e**. Yield 1.34 g (94%), orange crystals, mp > 300°C (decomp.). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 285.1098 (100) $[M + H]^+$. $C_{13}H_{12}N_6O_2^+$. *M* 285.1095.

2-R-6*H*-Benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*]-[1,2,4]triazin-6-ones 5a–5c and 2-*R*-[1,2,4]triazolo-[5',1':3,4][1,2,4]triazino[6,5-*c*]quinolin-6(7*H*)-ones 8a–8d (*general procedure*). A mixture of 4 mmol of azo coupling product **4a–4c** or **7a–7d** and 5 mL of DMF was heated under reflux for 15 min. The precipitate that formed after cooling was filtered off, washed with ethanol, and recrystallized from DMF.

6*H*-[1]Benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*]-[1,2,4]triazine-6-one (5a) was prepared using 1.03 g of compound **4a**. Yield 0.72 g (75%), light gray crystals, mp 278–280°C. ^{1}H NMR spectrum (DMSO-d_6, 500 MHz), δ, ppm: 7.73–7.76 m ($2H_{benz}$), 8.02–8.06 m ($1H_{benz}$), 9.43 s (1H, $CH_{triazole}$), 9.53–9.56 m ($1H_{benz}$). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 240.0515 (100) $[M + H]^+$. $C_{11}H_5N_5O_2^+$. *M* 240.0516.

2-Methyl-6*H*-[1]benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-one (5b) was prepared using 1.08 g of compound **4b**. Yield 1.05 g (91%), light gray crystals, mp 287–289°C. ^{1}H NMR spectrum (DMSO-d_6, 500 MHz), δ, ppm: 2.84 s (3H, CH_3), 7.71–7.75 m ($2H_{benz}$), 8.00–8.04 m (H_{benz}), 9.53 d.d ($1H_{benz}$, *J* 8.3, 1.7 Hz). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 254.0672 (100) $[M + H]^+$. $C_{12}H_7N_5O_2^+$. *M* 254.0673.

2-(Methylsulfanyl)-6*H*-[1]benzopyrano[4,3-*e*]-[1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-one (5c) was prepared using 1.21 g of compound **4c**. Yield 0.89 g (78%), light gray crystals, mp 295–297°C. ^{1}H NMR spectrum (DMSO-d_6, 500 MHz), δ, ppm: 2.90 s (3H, CH_3), 7.71–7.75 m ($2H_{benz}$), 8.00–8.04 m ($1H_{benz}$), 9.44 d.d ($1H_{benz}$, *J* 8.3, 1.7 Hz). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 286.0391 (100) $[M + H]^+$. $C_{12}H_7N_5O_2S^+$. *M* 286.0394.

[1,2,4]Triazolo[5',1':3,4][1,2,4]triazino[6,5-*c*]quinoline-6(7*H*)-one (8a) was prepared using 1.02 g of compound **7a**. Yield 0.80 g (71%), light gray crystals, mp > 300°C (decomp.). ^{1}H NMR spectrum (DMSO-d_6, 500 MHz), δ, ppm: 749–7.56 m ($2H_{benz}$), 7.86–7.89 m ($1H_{benz}$), 9.30 s ($1H_{triazole}$), 9.62 d ($1H_{benz}$, *J* 8.3 Hz), 12.62 s (1H, NH_{quin}). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 239.0675 (100) $[M + H]^+$. $C_{11}H_6N_6O^+$. *M* 239.0676.

2-Methyl[1,2,4]triazolo[5',1':3,4][1,2,4]triazino-[6,5-*c*]quinoline-6(7*H*)-one (8b) was prepared using 1.09 g of compound **7b**. Yield 0.74 g (73%), light gray crystals, mp > 300°C (decomp.). ^{1}H NMR spectrum (DMSO-d_6, 400 MHz), δ, ppm; 2.79 s (3H, CH_3), 7.46–7.52 m ($2H_{benz}$), 7.83–7.87 m ($1H_{benz}$), 9.60 d ($1H_{benz}$, *J* 8.3 Hz), 12.54 s (1H, NH_{quin}). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 253.0832 (100) $[M + H]^+$. $C_{12}H_8N_6O^+$. *M* 253.0833.

2-(Methylsulfanyl)[1,2,4]triazolo[5',1':3,4][1,2,4]-triazino[6,5-*c*]quinoline-6(7*H*)-one (8c) was prepared using 1.21 g of compound **7c**. Yield 1.16 g (82%), light gray crystals, mp > 300°C (decomp.). ^{1}H NMR spectrum (DMSO-d_6, 500 MHz), δ, ppm: 2.88 s (3H, SCH_3), 7.48–7.55 m ($2H_{benz}$), 7.84–7.87 m ($1H_{benz}$),

9.51 d (1H_{benz}, *J* 8.3 Hz), 12.49 s (1H, NH_{quin}). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 285.0555 (100) $[M + H]^+$. $C_{12}H_8N_6OS^+$. *M* 285.0553.

2-Ethyl[1,2,4]triazolo[5',1':3,4][1,2,4]triazino[6,5-*c*]quinoline-6(7*H*)-one (8d) was prepared using 1.14 g of compound **7e**. Yield 0.81 g (76%), light gray crystals, mp > 300°C (decomp.). ^{1}H NMR spectrum (DMSO-d_6, 400 MHz), δ, ppm: 1.47 t (3H, CH_3, *J* 7.6 Hz), 3.14 q (2H, CH_2, *J* 7.6 Hz), 7.49–7.55 m (2H_{benz}), 7.84–7.88 m (1H_{benz}), 9.63–9.66 m (1H_{benz}), 12.55 s (1H, NH_{quin}). Mass spectrum (ESI-HRMS), *m/z* (I_{rel}, %): 267.0990 (100) $[M + H]^+$. $C_{13}H_{10}N_6O^+$. *M* 267.0989.

CONCLUSIONS

A synthetic approach to the new tetracyclic systems benzopyrano[4,3-*e*][1,2,4]triazolo[5,1-*c*][1,2,4]triazin-6-one and [1,2,4]triazolo[5',1':3,4][1,2,4]-triazino[6,5-*c*]quinolin-6(7*H*)-one is proposed.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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